

Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102



Dr. Swati Rajput

Assistant Professor

Department of Electronics Engineering

Indian Institute of Technology (Indian School of Mines) Dhanbad

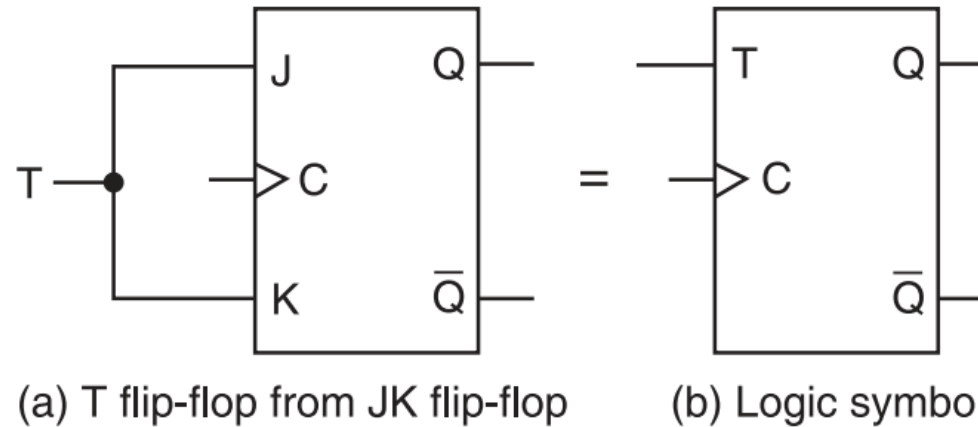
Digital Electronics: Key Topics at a Glance



Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	Combinational Logic Circuit: Adder, Subtractor, Magnitude Comparator, Code Converters etc. MSI Circuits: Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	Sequential Circuits: Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:	Reference Books:
1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.	1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015. 2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

T Flip Flop



Truth Table of T Flip Flop

CLK	T	Q_n	Q_{n+1}
1 (Triggered)	0	0	0
1 (Triggered)	0	1	1
1 (Triggered)	1	0	1
1 (Triggered)	1	1	0

Characteristics Table of T Flip Flop

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

$$Q_{n+1} = T \oplus Q_n$$

Excitation Table of T Flip Flop

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Characteristic Table of Flip Flops



Characteristics Table of SR Flip Flop

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

$$Q_{n+1} = S + Q_n \bar{R}$$

Characteristics Table of D Flip Flop

<u>Flop</u>		
D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

$$Q_{n+1} = D$$

Characteristics Table of T Flip Flop

<u>Flop</u>		
T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

$$Q_{n+1} = T \oplus Q_n$$

Characteristics Table of JK Flip Flop

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

Excitation Table of Flip Flops



Excitation Table of SR Flip Flop

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Excitation Table of D Flip Flop

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Excitation Table of T Flip Flop

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Flip Flop Conversion



Steps:

1. Identify the Required and Available Flip Flop
2. Make Characteristic Table for Required Flip Flop
3. Make Excitation table for the Available Flip Flop
4. Write Boolean Expression for the Available Flip Flop
5. Draw the Circuit.

JK FLIP Flop to D Flip Flop

Step 1:

Available Flip Flop is JK Flip Flop
Required Flip Flop is D Flip Flop

Step 2:

Characteristic Table of D Flip Flop

Characteristics Table of D Flip Flop

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

$$Q_{n+1} = D$$

Step 3:

Excitation Table of Available Flip Flop

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Flip Flop Conversion

JK FLIP Flop to D Flip Flop

Step 2:

Characteristic Table of D Flip Flop

Characteristics Table of D Flip Flop

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Step 3:

Excitation Table of Available Flip Flop

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

External Input	Present State	Next State	Flip-flop Inputs	
D	Q_n	Q_{n+1}	J	K
0	0	0	0	x
0	1	0	x	1
1	0	1	1	x
1	1	1	x	0

(a) Conversion table

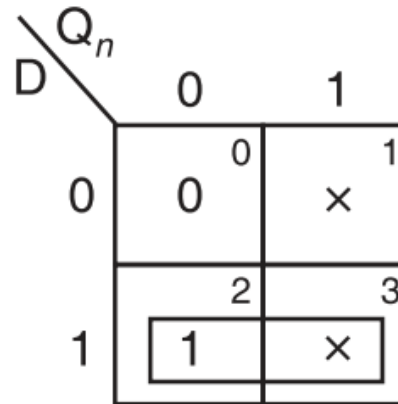
Flip Flop Conversion

JK FLIP Flop to D Flip Flop

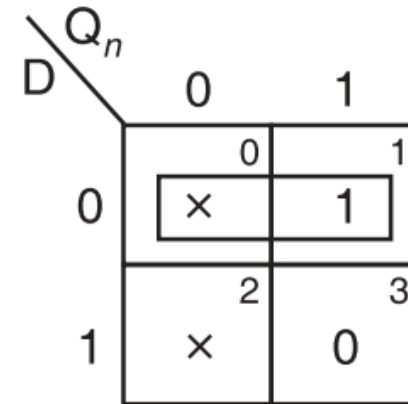
External Input	Present State	Next State	Flip-flop Inputs	
D	Q_n	Q_{n+1}	J	K
0	0	0	0	x
0	1	0	x	1
1	0	1	1	x
1	1	1	x	0

(a) Conversion table

Step 4: Write Boolean Expression for the Available Flip Flop



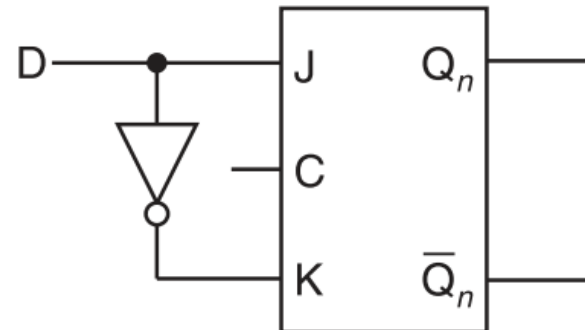
$$J = D$$



$$K = \bar{D}$$

(b) K-maps for J and K

Step 5: Draw the Circuit.



(c) Logic diagram

Flip Flop Conversion

Steps:

1. Identify the Required and Available Flip Flop
2. Make Characteristic Table for Required Flip Flop
3. Make Excitation table for the Available Flip Flop
4. Write Boolean Expression for the Available Flip Flop
5. Draw the Circuit.

D FLIP Flop to JK Flip Flop

Step 1:

Available Flip Flop is D Flip Flop
Required Flip Flop is JK Flip Flop

Step 2:

Characteristic Table of JK Flip Flop

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Step 3:

Excitation Table of Available Flip Flop

Excitation Table of D Flip Flop

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Flip Flop Conversion

D FLIP Flop to JK Flip Flop

Step 2:

Characteristic Table of JK Flip Flop

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Step 3:

Excitation Table of Available Flip Flop

Excitation Table of D Flip Flop

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

External Inputs		Present State	Next State	Flip-flop Input
J	K	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

(a) Conversion table

Flip Flop Conversion



D FLIP Flop to JK Flip Flop

External Inputs		Present State	Next State	Flip-flop Input
J	K	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

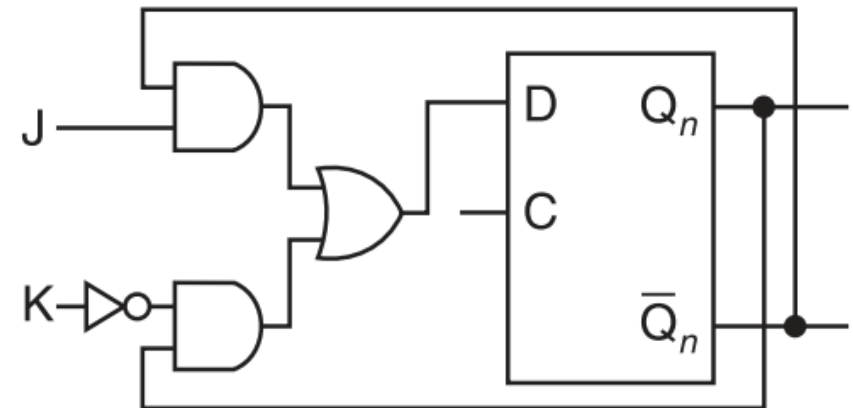
(a) Conversion table

Step 4: Write Boolean Expression for the Available Flip Flop

J \ KQ _n				
	00	01	11	10
0	0 ⁰	1 ¹	0 ³	0 ²
1	1 ⁴	1 ⁵	0 ⁷	1 ⁶

$$D = J\bar{Q}_n + \bar{K}Q_n$$

(b) K-maps for D



(c) Logic diagram

Step 5: Draw the Circuit.

Flip Flop Conversion

Steps:

1. Identify the Required and Available Flip Flop
2. Make Characteristic Table for Required Flip Flop
3. Make Excitation table for the Available Flip Flop
4. Write Boolean Expression for the Available Flip Flop
5. Draw the Circuit.

SR FLIP Flop to JK Flip Flop

Step 1:

Available Flip Flop is SR Flip Flop
Required Flip Flop is JK Flip Flop

Step 2:

Characteristic Table of JK Flip Flop

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Step 3:

Excitation Table of Available Flip Flop

Excitation Table of SR Flip Flop

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Flip Flop Conversion



SR FLIP Flop to JK Flip Flop

Step 2:

Characteristic Table of JK Flip Flop

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Step 3:

Excitation Table of Available Flip Flop

Excitation Table of SR Flip Flop

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

External Inputs		Present State	Next State	Flip-flop Inputs	
J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	x
0	0	1	1	x	0
0	1	0	0	0	x
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	x	0
1	1	0	1	1	0
1	1	1	0	0	1

(a) Conversion table

Flip Flop Conversion



SR FLIP Flop to JK Flip Flop

External Inputs		Present State	Next State	Flip-flop Inputs	
J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	x
0	0	1	1	x	0
0	1	0	0	0	x
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	x	0
1	1	0	1	1	0
1	1	1	0	0	1

(a) Conversion table

Step 4: Write Boolean Expression for the Available Flip Flop

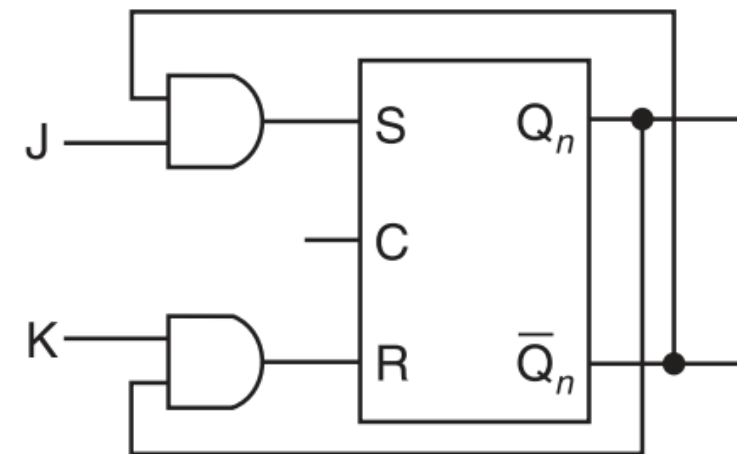
J	KQ_n			
	00	01	11	10
0	0 ⁰ 0	x ¹ x	0 ³ 0	0 ² 0
1	1 ⁴ 1	x ⁵ x	0 ⁷ 0	1 ⁶ 1

$$S = J\bar{Q}_n$$

J	KQ_n			
	00	01	11	10
0	x ⁰ x	0 ¹ 0	1 ³ 1	x ² x
1	0 ⁴ 0	0 ⁵ 0	1 ⁷ 1	0 ⁶ 0

$$R = KQ_n$$

(b) K-maps for S and R



(c) Logic diagram

Step 5: Draw the Circuit.

Flip Flop Conversion

Steps:

1. Identify the Required and Available Flip Flop
2. Make Characteristic Table for Required Flip Flop
3. Make Excitation table for the Available Flip Flop
4. Write Boolean Expression for the Available Flip Flop
5. Draw the Circuit.

JK FLIP Flop to SR Flip Flop

Step 1:

Available Flip Flop is JK Flip Flop
Required Flip Flop is SR Flip Flop

Step 2:

Characteristic Table of SR Flip Flop

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

Step 3:

Excitation Table of Available Flip Flop

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Flip Flop Conversion

JK FLIP Flop to SR Flip Flop

Step 2:

Characteristic Table of SR Flip Flop

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

Step 3:

Excitation Table of Available Flip Flop

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

External Inputs		Present State	Next State	Flip-flop Inputs	
S	R	Q_n	Q_{n+1}	J	K
0	0	0	0	0	x
0	0	1	1	x	0
0	1	0	0	0	x
0	1	1	0	x	1
1	0	0	1	1	x
1	0	1	1	x	0

(a) Conversion table

Flip Flop Conversion

JK FLIP Flop to SR Flip Flop

External Inputs		Present State	Next State	Flip-flop Inputs	
S	R	Q_n	Q_{n+1}	J	K
0	0	0	0	0	x
0	0	1	1	x	0
0	1	0	0	0	x
0	1	1	0	x	1
1	0	0	1	1	x
1	0	1	1	x	0

(a) Conversion table

Step 4: Write Boolean Expression for the Available Flip Flop

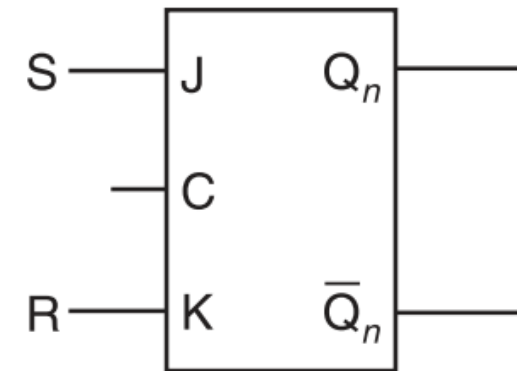
S	RQ_n			
	00	01	11	10
0	0 ⁰	x ¹	x ³	0 ²
1	1 ⁴	x ⁵	x ⁷	x ⁶

$$J = S$$

S	RQ_n			
	00	01	11	10
0	x ⁰	0 ¹	1 ³	x ²
1	x ⁴	0 ⁵	x ⁷	x ⁶

$$K = R$$

(b) K-maps for J and K



(c) Logic diagram

Step 5: Draw the Circuit.

State Diagram



- ❖ The state diagram or state graph is a pictorial representation of the relationships between the present state, the input, the next state, and the output of a sequential circuit, i.e. the state diagram is a pictorial representation of the behavior of a sequential circuit

D Flip Flop

Characteristics Table of D Flip Flop

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

$$Q_{n+1} = D$$

Excitation Table of D Flip Flop

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$D(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

State Diagram



D Flip Flop

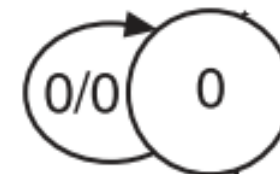
Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$D(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.

❖ Let $S_1 = 0$ and $S_2 = 1$

❖ **First Case when Present state $S_1 = 0$ and Input is 0 → The next state is 0 itself.**

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$D(t)$	$Q(t+1)$
0	0	0



State Diagram



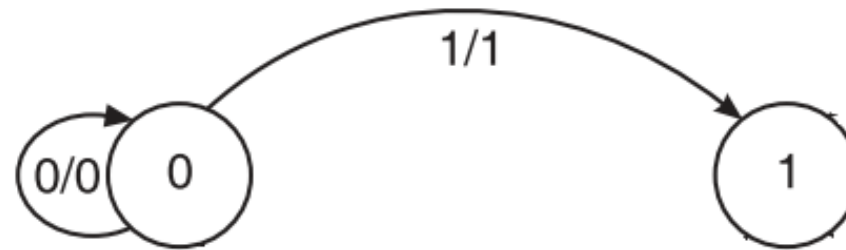
D Flip Flop

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$D(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.

❖ Let $S_1 = 0$ and $S_2 = 1$

❖ **Second Case when Present state $S_1 = 0$ and Input is 1 → The next state is 1**



State Diagram



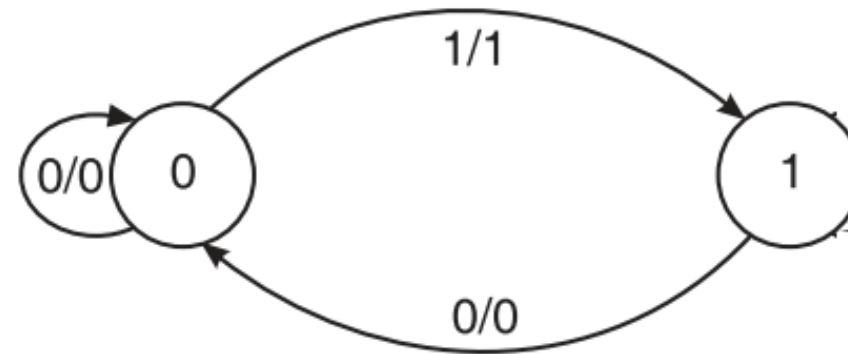
D Flip Flop

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$D(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.

❖ Let $S_1 = 0$ and $S_2 = 1$

❖ **Third Case when Present state $S_2 = 1$ and Input is 0 → The next state is 0**



State Diagram



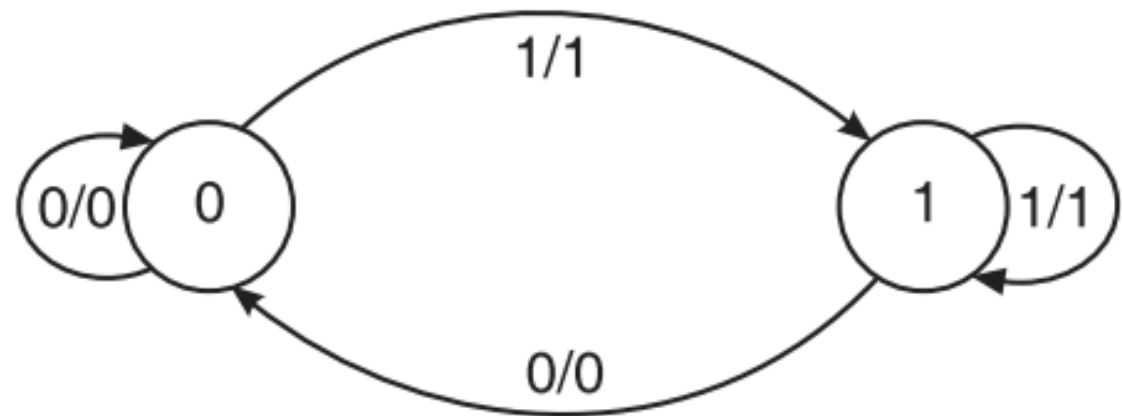
D Flip Flop

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$D(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.

❖ Let $S_1 = 0$ and $S_2 = 1$

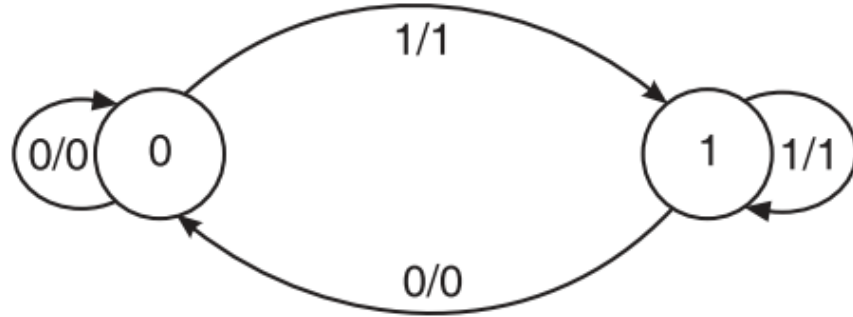
❖ **Fourth Case when Present state $S_2 = 1$ and Input is 1 → The next state is 1**



State Diagram



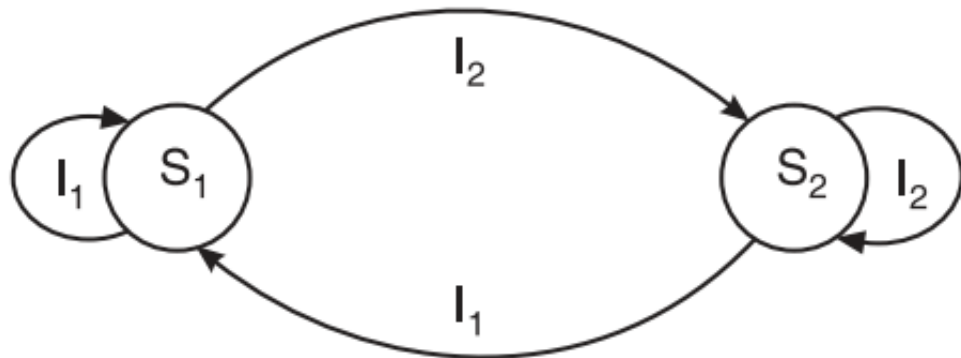
D Flip Flop



(b) State diagram (Mealy model)

PS	N/S, O/P	
	D = 0	D = 1
0	0, 0	1, 1
1	0, 0	1, 1

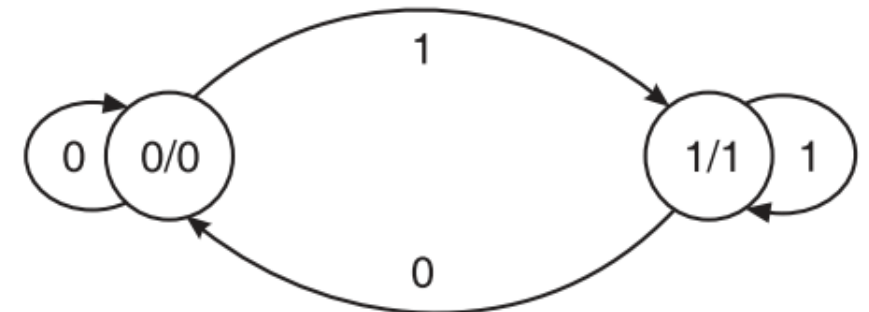
(c) State table



(d) General state diagram

		D(t)	
		0	1
Q(t)	0	0	1
	1	2	3

(e) K-map for $Q(t+1)$



(f) Moore model

State Diagram



T Flip Flop

Characteristics Table of T Flip

Flop

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

$$Q_{n+1} = T \oplus Q_n$$

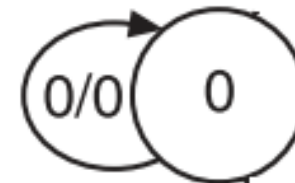
Excitation Table of T Flip Flop

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$T(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

(a) Excitation table

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$T(t)$	$Q(t+1)$
0	0	0



- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **First Case when Present state $S_1 = 0$ and Input is 0 → The next state is 0 itself.**

State Diagram

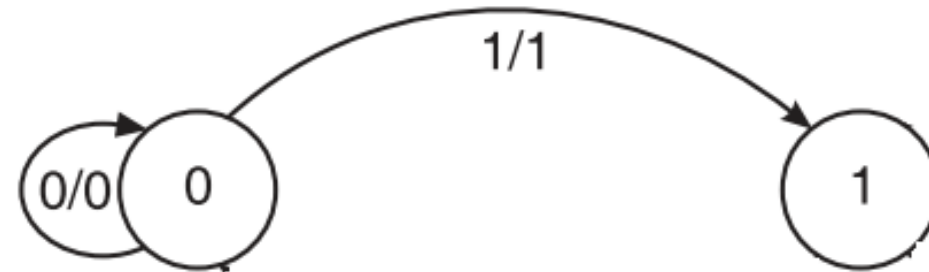


T Flip Flop

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$T(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **Second Case when Present state $S_1 = 0$ and Input is 1 → The next state is 1**



State Diagram

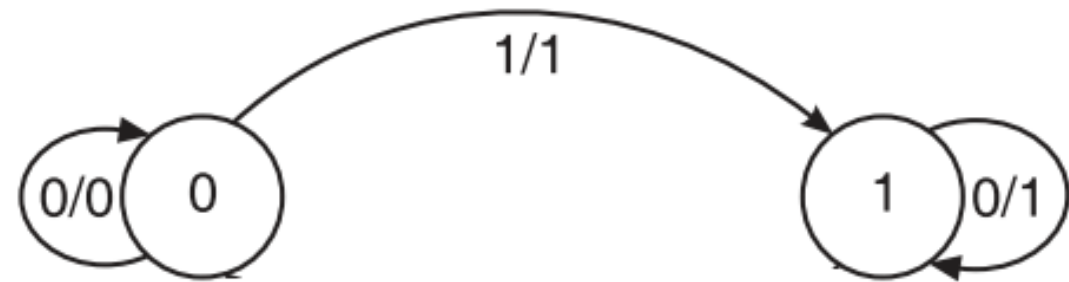


T Flip Flop

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$T(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **Second Case when Present state $S_1 = 1$ and Input is 0 → The next state is 1**



State Diagram

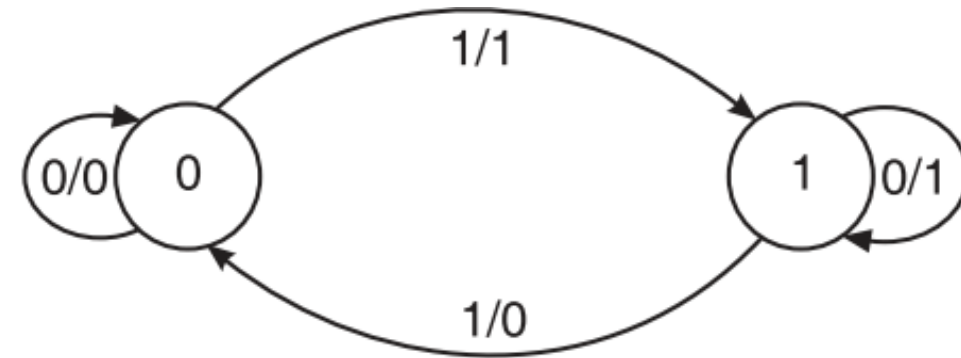


T Flip Flop

Present state (PS)	Input to FF	Next state (NS)
$Q(t)$	$T(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

(a) Excitation table

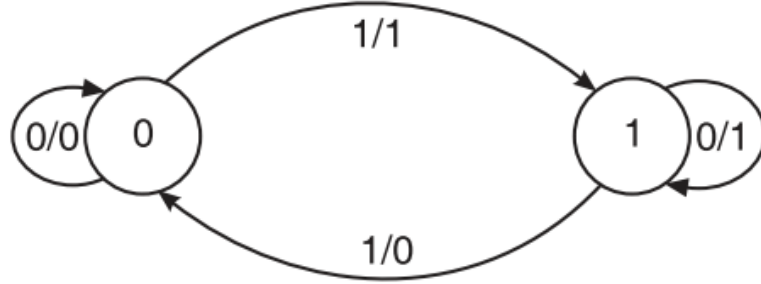
- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **Second Case when Present state $S_1 = 1$ and Input is 1 → The next state is 0**



State Diagram



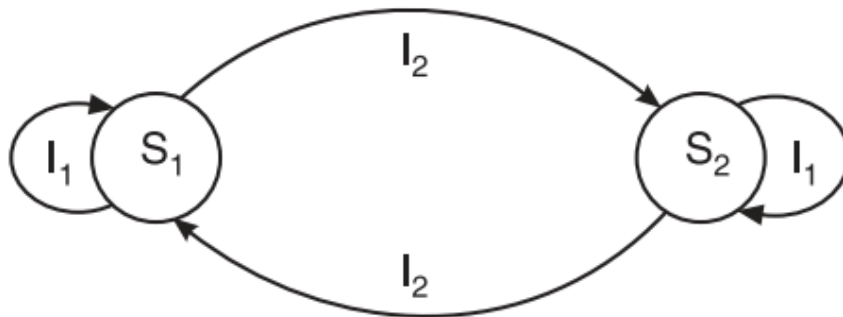
T Flip Flop



(b) State diagram (Mealy model)

PS	N/S, O/P	
	T = 0	T = 1
0	0, 0	1, 1
1	1, 1	0, 0

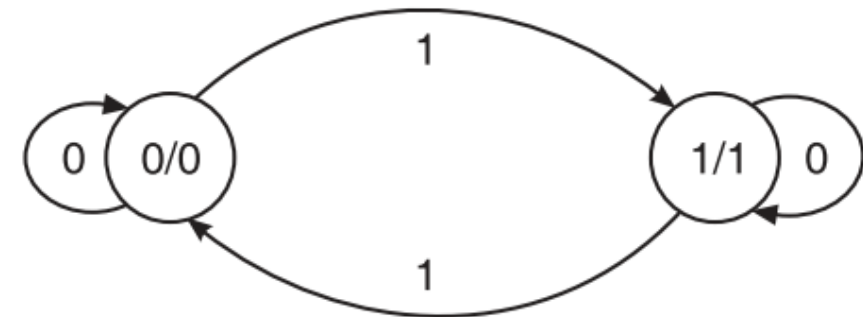
(c) State table



(d) General state diagram

		T(t)	
		0	1
Q(t)	0	0	1
	1	1	2

(e) K-map for $Q(t+1)$



(f) Moore model

State Diagram



SR Flip Flop

Characteristics Table of SR Flip Flop

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

Excitation Table of SR Flip Flop

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	S(t)	R(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
1	0	0	1
1	0	1	0
1	1	0	1

(a) Excitation table

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	S(t)	R(t)	Q(t+1)
0	0	0	0
0	0	1	0

$$Q_{n+1} = S + Q_n \bar{R}$$

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_1 = 0$ and Inputs are $S = 0$ and $R = 0 \rightarrow$ The next state is 0 itself.**



State Diagram

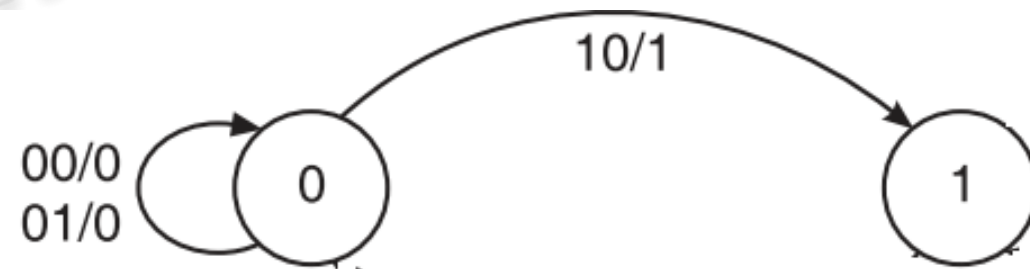


SR Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	S(t)	R(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
1	0	0	1
1	0	1	0
1	1	0	1

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_1 = 0$ and Inputs are $S = 1$ and $R = 0 \rightarrow$ The next state is 1.**



State Diagram

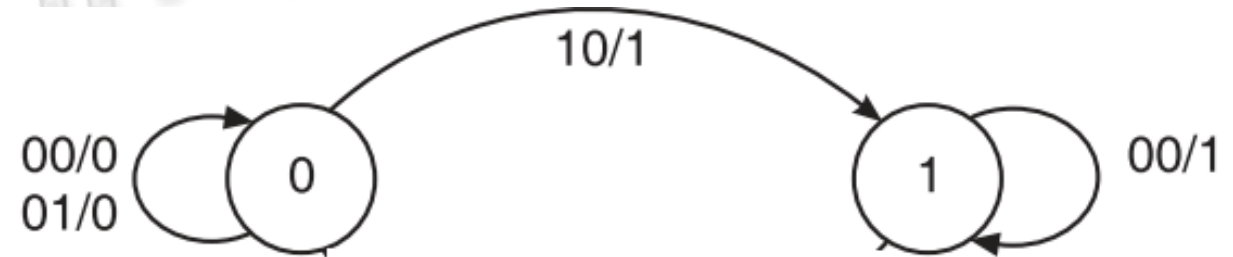


SR Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	S(t)	R(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
1	0	0	1
1	0	1	0
1	1	0	1

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_2 = 1$ and Inputs are $S = 0$ and $R = 0 \rightarrow$ The next state is 1.**



State Diagram

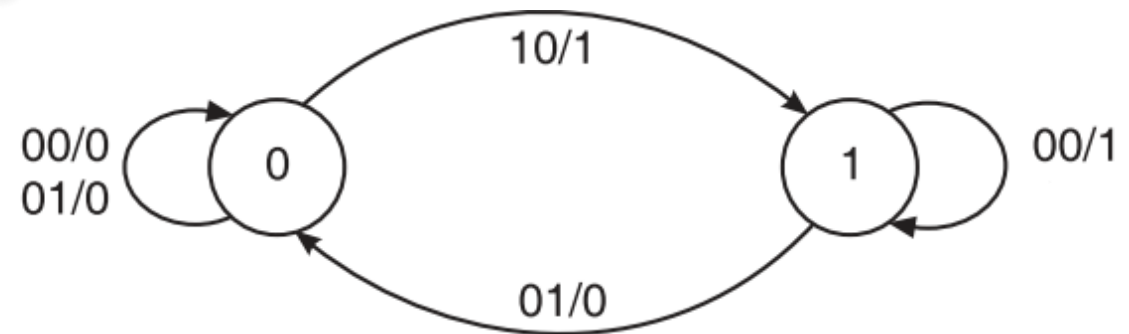


SR Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	S(t)	R(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
1	0	0	1
1	0	1	0
1	1	0	1

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_2 = 1$ and Inputs are $S = 0$ and $R = 1$ → The next state is 0.**



State Diagram

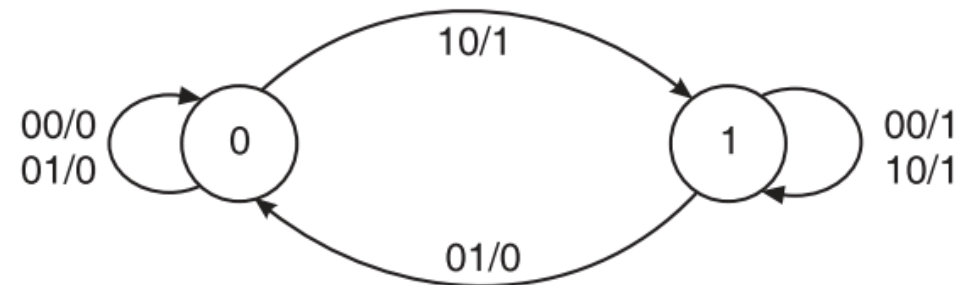


SR Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	S(t)	R(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
1	0	0	1
1	0	1	0
1	1	0	1

(a) Excitation table

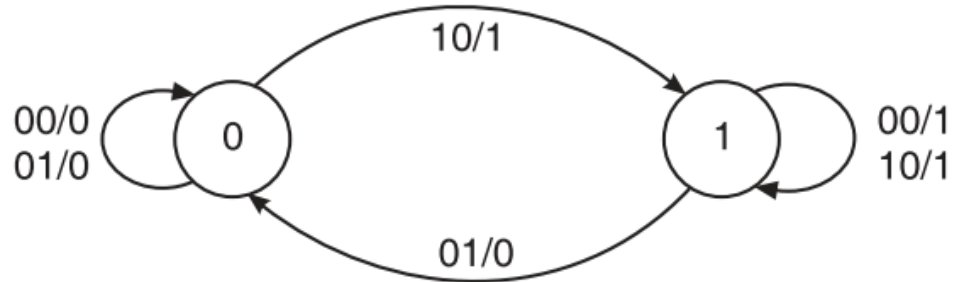
- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_2 = 1$ and Inputs are $S = 1$ and $R = 0 \rightarrow$ The next state is 1.**



State Diagram



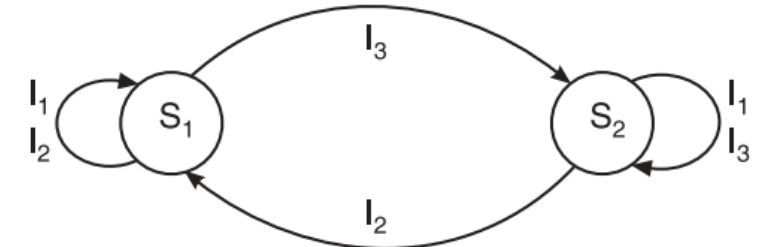
SR Flip Flop



(b) State diagram (Mealy model)

PS	NS, O/P		
	S R	S R	S R
0	0, 0	0, 0	1, 1
1	1, 1	0, 0	1, 1

(c) State table

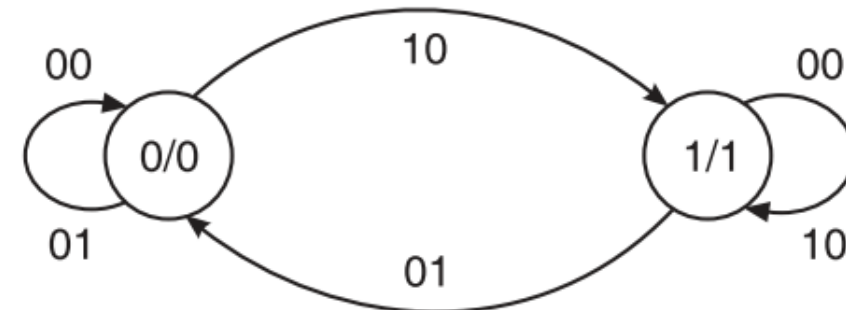


(d) General state diagram

$S(t) R(t)$		$Q(t)$			
		00	01	11	10
$Q(t)$	0	0	0	×	1
	1	1	0	×	1

(e) K-map for $Q(t+1)$

$$Q(t+1) = Q(t) \bar{R}(t) + S(t)$$



(f) Moore model

State Diagram



JK Flip Flop

Characteristics Table of JK Flip Flop

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.

❖ Let $S_1 = 0$ and $S_2 = 1$

❖ **When Present state $S_1 = 0$ and Inputs are $S = 0$ and $R = 0 \rightarrow$ The next state is 0 itself.**

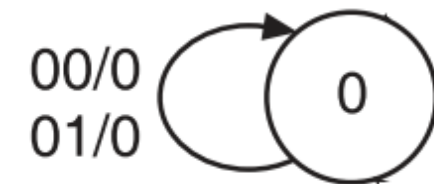
Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Present state (PS)	Inputs to FF		Next state (NS)
$Q(t)$	J(t)	K(t)	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a) Excitation table

Present state (PS)	Inputs to FF		Next state (NS)
$Q(t)$	J(t)	K(t)	$Q(t+1)$
0	0	0	0
0	0	1	0



State Diagram

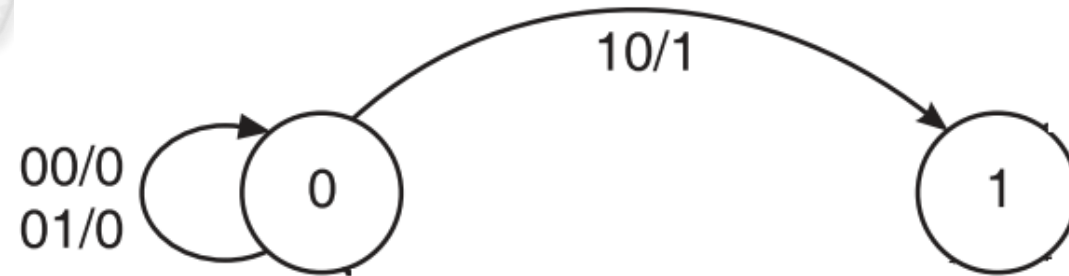


JK Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	J(t)	K(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_1 = 0$ and Inputs are $J = 1$ and $K = 0 \rightarrow$ The next state is 1.**



State Diagram

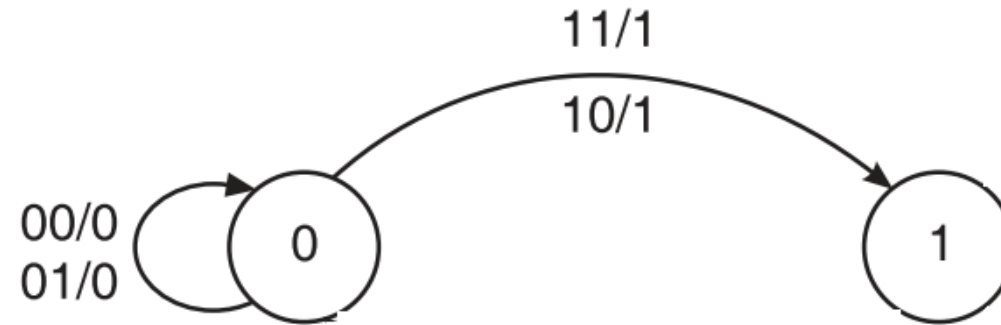


JK Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
$Q(t)$	$J(t)$	$K(t)$	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_1 = 0$ and Inputs are $J = 1$ and $K = 1 \rightarrow$ The next state is 1.**



State Diagram

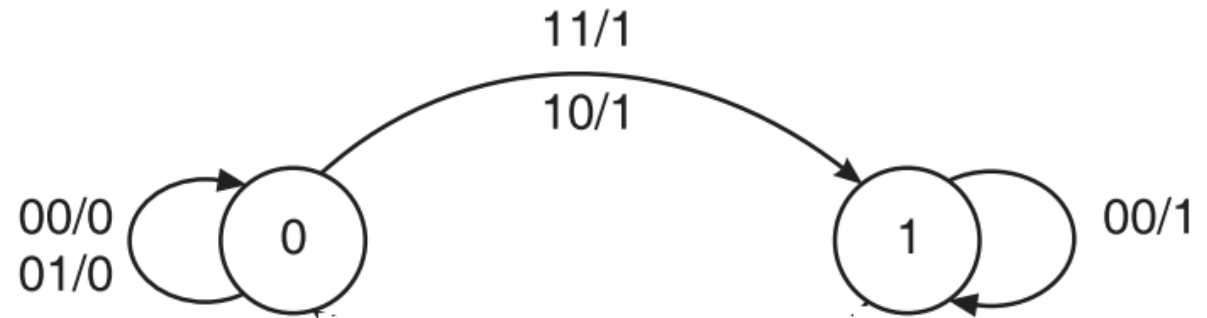


JK Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	J(t)	K(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_2 = 1$ and Inputs are $J = 0$ and $K = 0 \rightarrow$ The next state is 1.**



State Diagram

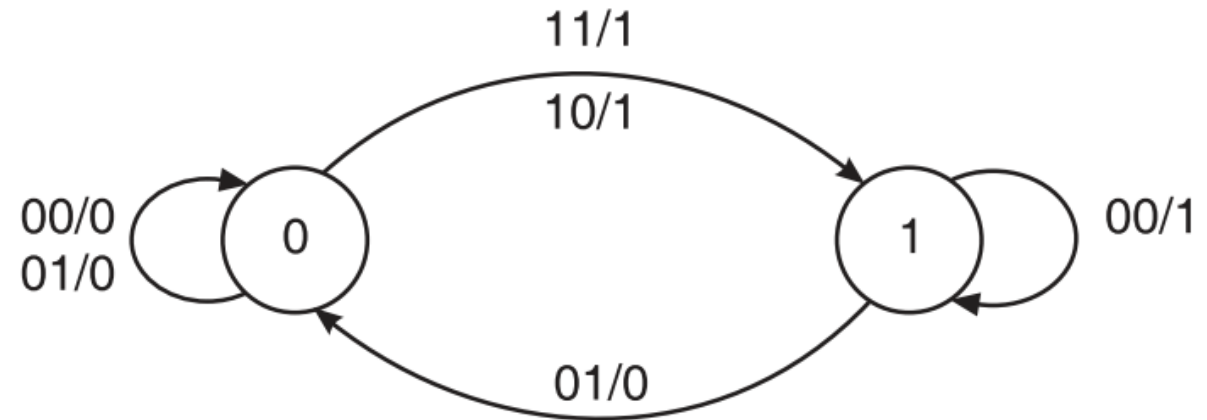


JK Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
$Q(t)$	$J(t)$	$K(t)$	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_2 = 1$ and Inputs are $J = 0$ and $K = 1 \rightarrow$ The next state is 0.**



State Diagram

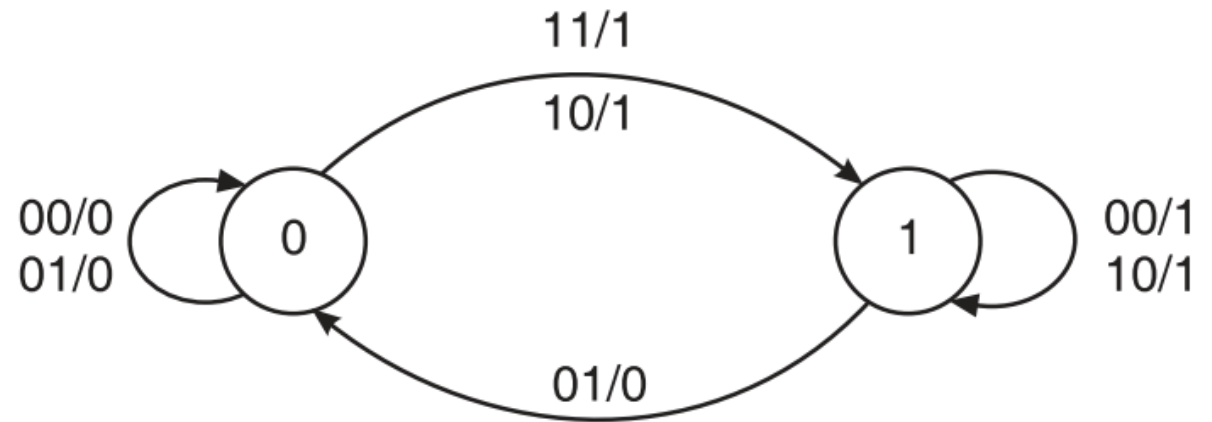


JK Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	J(t)	K(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a) Excitation table

- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_2 = 1$ and Inputs are $J = 1$ and $K = 0 \rightarrow$ The next state is 1.**



State Diagram

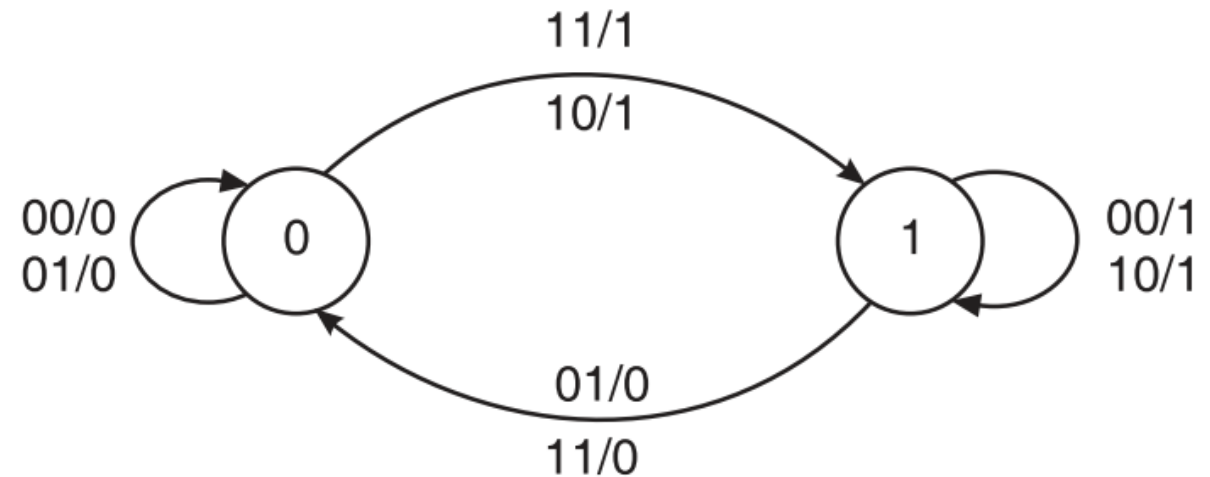


JK Flip Flop

Present state (PS)	Inputs to FF		Next state (NS)
Q(t)	J(t)	K(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a) Excitation table

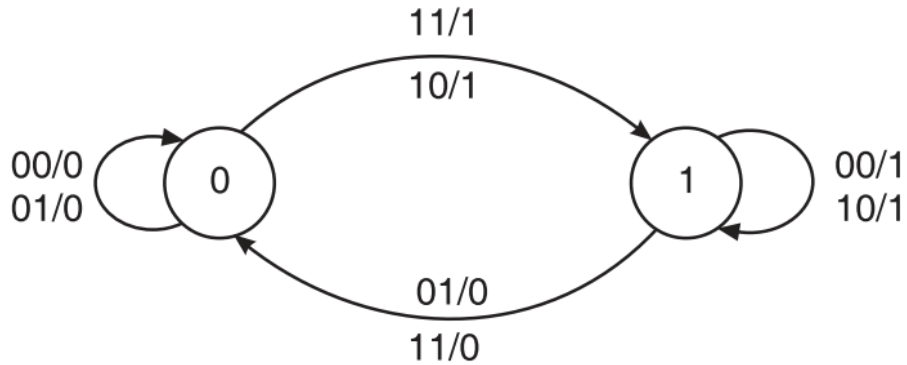
- ❖ S_1 and S_2 represent the states of the FF and I_1 and I_2 correspond to the input conditions.
- ❖ Let $S_1 = 0$ and $S_2 = 1$
- ❖ **When Present state $S_2 = 1$ and Inputs are $J = 1$ and $K = 1 \rightarrow$ The next state is 0.**



State Diagram



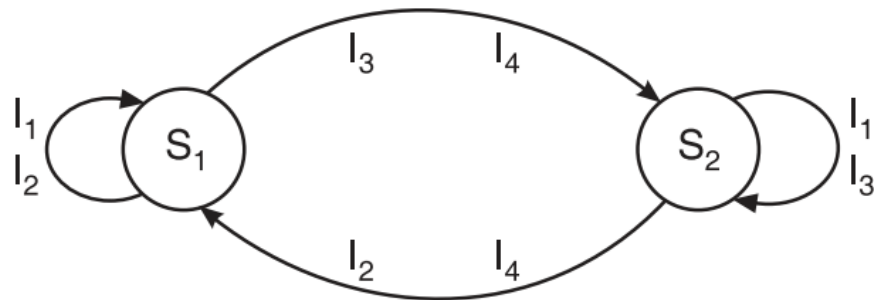
JK Flip Flop



(b) State diagram (Mealy model)

PS	NS, O/P			
	JK			
	0 0	0 1	1 0	1 1
0	0, 0	0, 0	1, 1	1, 1
1	1, 1	0, 0	1, 1	0, 0

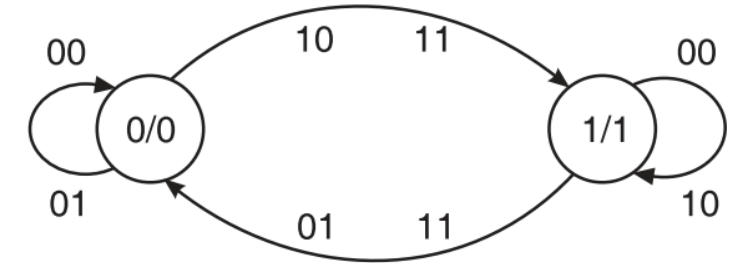
(c) State table



(d) General state diagram

J(t) K(t)		Q(t)			
		00	01	11	10
Q(t)	0	0 0	1 0	3 1	2 1
	1	4 1	5 0	7 0	6 1

(e) K-map for $Q(t+1)$



(f) Moore model

$$Q(t+1) = J(t) \bar{Q}(t) + Q(t) \bar{K}(t)$$